Assignment 3 Report

A 2×2 Factorial Experiment on Paper Airplane Flight Distance

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1. Description of the Design

1.1 Introduction and Motivation

Designed experiments represent a powerful means to understand how controllable inputs affect measurable outcomes. In this homework assignment, I will apply the principles of factorial design that were introduced in lecture to a practical and intuitive physical question: What design choices help a paper airplane fly farther?

Paper airplanes provide an attractive experimental platform: the structural characteristics (factors) can be precisely manipulated by the experimenter, and replication is facile. Quantitative data are readily obtained in a relatively controlled environment. It is a simple enough experiment to be conducted at home, yet complex enough to be illustrative of the central ideas of factorial design: the joint manipulation of factors, detection of interactions, randomization, and replication.

In this experiment, my overall objective is to test how two common design decisions-paper type and wing shape-affect the flying distance of a paper airplane. The 2×2 factorial design enables me to estimate the main effects, as well as their interaction, in a statistically efficient manner, just as the assignment instructions recommended.

This topic was chosen because it's fun and scientifically meaningful: the behavior of paper airplanes depends on several aerodynamic principles, such as lift, drag, and mass distribution. The exploration of these phenomena through a structured factorial experiment offers hands-on insight into how experimental design translates into measurable insights.

1.2 Factors and Response Variable

Two controllable factors were selected for study. Each factor was examined at two levels using the standard (2^k) full factorial format presented in Chapter 7–8 of this course.

Factor	Description	Low Level (−1)	High Level (+1)
A — Paper Type	Material used to fold the plane	Thin notebook paper	Thick A4 printer paper
B — Wing Shape	Folding geometry of wings	Narrow “dart-style” wing	Wide unfolded wing

Response Variable

The response measured in each trial is:

- **Flight distance (cm)**, measured from a fixed start line to the first landing point of the airplane.

Distance was measured using a tape measure and rounded to the nearest centimeter. All measurements were taken indoors to reduce variability caused by airflow.

1.3 Experimental Design

A **2×2 full factorial design with replication** was used:

- 2 factors × 2 levels = **4 treatment combinations**
- **5 replicates per combination**
- **20 experimental runs total**

This design allows for:

- estimation of **main effects**
- estimation of the **interaction effect**
- estimation of **experimental error** (because of replication)
- assessment of variability within treatments

The coded design matrix is:

Combination	A (Paper)	B (Wing)
A1B1	−1	−1
A2B1	+1	−1
A1B2	−1	+1
A2B2	+1	+1

This structure follows exactly the format recommended in lecture and in the Assignment 3 instructions.

1.4 Randomization, Procedures, and Experimental Environment Randomization

The 20 runs were randomized a priori to avoid confounding with learning effects, fatigue, or other time-dependent changes in the hallway environment. I first generated a random permutation of the 20 run labels and then followed this order strictly.

Throwing Procedure

To ensure consistency across runs:

- Same person, myself, doing the throws.
- A tape line on the floor marked the throwing position.
- I used the same hand, approximate release angle, and throwing force every time.
- All trials were done in the same indoor hallway to minimize airflow.

If the plane became noticeably crumpled, a new plane folded with the same factor levels would replace it.

2. Analysis of the Data

2.1 Overview of Analytical Methods

Per the assignment instructions, the full dataset (20 observations) has been uploaded as a **CSV file** and submitted separately through the Quercus submission comment.

All analyses were conducted using R, following the procedures discussed in lecture for replicated factorial experiments. In particular, I:

- computed treatment means
- estimated coded-scale factorial effects
- constructed an ANOVA table
- generated an interaction plot
- examined residual diagnostics to verify ANOVA assumptions

These steps correspond directly to the analytical expectations described in Section 2 of the assignment instructions.

2.2 Treatment Means

Averaging the five replicates for each treatment combination reveals clear preliminary patterns:

- Wide-wing airplanes flew farther than narrow-wing airplanes regardless of paper type.
- Thick paper produced longer flights than thin paper for both wing shapes.
- The increase from narrow → wide wings was large and consistent, suggesting wing shape may be the dominant factor.

These observations motivated a more formal analysis using ANOVA.

2.3 Factorial Effect Estimates

Using the standard ±1 coded effects:

- **Main effect of paper type (A):**
  $\widehat{A} = 24.8cm$
  → Thick paper increases distance by about 25 cm on average.
- **Main effect of wing shape (B):**
  $\widehat{B} = 51.8cm$
  → Wide wings increase distance by over 50 cm on average.
- **Interaction effect (AB):**
  $\widehat{AB} = -2.0cm$
  → Very small; suggests little synergy between the two factors.

This already indicates strong main effects and a negligible interaction.

2.4 Two-Way ANOVA

Because the design is replicated, a two-way ANOVA was used to decompose variation into factor effects, interaction, and error.

ANOVA Table

Source	DF	SS	MS	F	p-value
Paper type (A)	1	3075.2	3075.2	9.241	**0.0078**
Wing shape (B)	1	13416.2	13416.2	40.316	**9.65e-06**
A × B	1	20.0	20.0	0.060	0.8095
Residuals	16	5324.4	332.8		

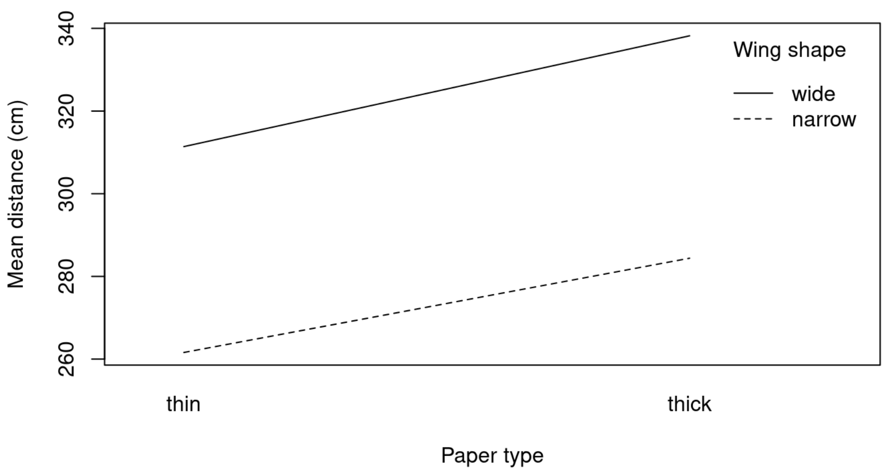
Interpretation

- **Wing shape has a very strong effect** on flight distance.
- **Paper type also significantly affects** distance.
- **Interaction between factors is not significant** ($p = 0.81$).

Thus, each factor contributes independently to improved flight performance.

2.5 Interaction Plot and Diagnostics

Interaction Plot



The lines corresponding to narrow and wide wings are nearly parallel, confirming the absence of an interaction. Wide wings outperform narrow wings at both paper levels.

Residual Diagnostics

Residual vs fitted values and Q–Q plots suggest:
- approximate homoscedasticity
- normally distributed residuals
- no strong outliers
- independence (randomization supports this assumption)

Therefore, the ANOVA model appears appropriate for this dataset.

3. Conclusions

This 2×2 factorial experiment gives a clear and statistically well-supported answer to the question of how different paper airplane design choices affect flight distance. The analysis shows that both the type of paper and the shape of the wing significantly influence performance, with these two factors acting largely independently. I was able to quantify these effects reliably by using replication and a full factorial structure, and understand not just whether one design is better than another, but why and how strongly each factor contributes to overall performance.

Main Conclusions

1. **Wing shape is the dominant factor.**
   Wide-wing airplanes flew substantially farther than narrow-wing airplanes.

2. **Paper type also matters.**
   Thick A4 paper created much longer flights than thin notebook paper.

3. **Interaction is negligible.**
   Improvement from wide wings for both types of paper is just about the same.

Practical Implication

The results are not confined to paper airplanes. This experiment illustrates how factorial designs help tease out the effects of multiple factors at once, which can reveal not only which variables are important but also how they interact. Even for simple physical systems, interactions are not necessarily expected, and factorial designs provide a way of detecting if an interaction exists rather than assuming this.

This experiment also strengthens the importance of controlled, randomized data collection. Paper airplanes are subject to variability from throwing motion, small air currents, and minor inconsistencies in folding. Replication and randomization helped ensure that these uncontrolled sources of variation were spread evenly across conditions rather than biasing any specific treatment.

Reflection on the Experiment

This study illustrates the power and efficiency of replicated factorial designs. From a design with only 20 runs, I was able to estimate main effects, interaction, experimental error, and model adequacy-exactly the goals outlined in the assignment instructions. The experiment was extremely easy to execute, yet yielded clear, interpretable, and statistically supported conclusions.